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Measure Theory

1.1 Outer Measure

Definition 1.1.1 ▸ Outer Measure

A function φ defined for every subset A of an arbitrary set X is called an ***outer measure*** on X if the following conditions are satisfied:

(i) $\varphi(\emptyset) = 0$,
(ii) $0 \leq \varphi(A) \leq \infty$ whenever $A \subset X$,
(iii) $\varphi(A_1) \leq \varphi(A_2)$ whenever $A_1 \subset A_2$,
(iv) $\varphi(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \varphi(A_i)$ for every countable collection of sets $\{A_i\}$ in X.

Example 1.1.2 ▸ Outer Measure

(i) In an arbitrary set X, define $\varphi(A) = 1$ if A is nonempty and $\varphi(\emptyset) = 0$.
(ii) Let $\varphi(A)$ be the number (possibly infinite) of points in A.
(iii) Let $\varphi(A) = \begin{cases} 0 & \text{if card } A \leq \aleph_0, \\ 1 & \text{if card } A > \aleph_0. \end{cases}$
(iv) If X is a metric space, fix $\varepsilon > 0$. Let $\varphi(A)$ be the smallest number of balls of radius ε that cover A.
(v) Select a fixed x_0 in an arbitrary set X, and let

$$\varphi(A) = \begin{cases} 0 & \text{if } x_0 \notin A, \\ 1 & \text{if } x_0 \in A; \end{cases}$$

φ is called the ***Dirac measure*** concentrated at x_0.

Definition 1.1.3

Let φ be an outer measure on a set X. A set $E \subset X$ is called φ***-measurable*** if

$$\varphi(A) = \varphi(A \cap E) + \varphi(A - E)$$

for every set $A \subset X$. In view of property (iv) above, observe that φ-measurability requires only

$$\varphi(A) \geq \varphi(A \cap E) + \varphi(A - E).$$

Lemma 1.1.4

A set $E \subset X$ is φ-measurable if and only if

$$\varphi(P \cup Q) = \varphi(P) + \varphi(Q)$$

for all sets P and Q such that $P \subset E$ and $Q \subset E^c$.

Proof. For Necessity, if E is φ-measurable , then for all such P, Q,

$$\varphi\left(P \cup Q\right) = \varphi\left(\left(P \cup Q\right) \cap E\right) + \varphi\left(\left(P \cup Q\right) - E\right) = \varphi\left(P\right) + \varphi\left(Q\right)$$

For sufficiency , let $P = A \cap E$, $Q = A \cap E^c = A \setminus E$. Then

$$\varphi\left(A\right) = \varphi\left(P \cup Q\right) = \varphi\left(P\right) + \varphi\left(Q\right) = \varphi\left(A \cap E\right) + \varphi\left(A \setminus E\right)$$

□

Theorem 1.1.5

Suppose φ is an outer measure on an arbitrary set X. Then the following four statements hold:

(i) E is φ-measurable whenever $\varphi(E) = 0$.

(ii) $\varnothing$ and X are φ-measurable.

(iii) $E_1 - E_2$ is φ-measurable whenever E_1 and E_2 are φ-measurable.

(iv) If $\{E_i\}$ is a ***countable collection of disjoint φ-measurable sets***, then $\bigcup_{i=1}^{\infty} E_i$ is φ-measurable and

$$\varphi\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \varphi(E_i).$$

More generally, if $A \subset X$ is an arbitrary set, then

$$\varphi(A) = \sum_{i=1}^{\infty} \varphi(A \cap E_i) + \varphi(A \cap \tilde{S}),$$

where
$$S = \bigcup_{i=1}^{\infty} E_i.$$

Proof. 1. Take any $A \subseteq X$. Since $A \cap E \subseteq E$, $\varphi(A \cap E) = 0$. Thus
$$\varphi(A \cap E) + \varphi(A \cap E^c) = \varphi(A \cap E^c) \leq \varphi(A)$$

2. Follows immediately frome lemma 1.1.4.
3. Take any P, Q such that $P \subseteq (E_1 \setminus E_2) = E_1 \cap E_2^c$, $Q \subseteq (E_1 \setminus E_2)^c = E_1^c \cup E_2$.
$$\varphi(Q) = \varphi(Q \cap E_2) + \varphi(Q - E_2)$$
Note that $Q - E_2 \subseteq E_1^c$, $Q \cap E_2 \subseteq E_2$. Since $P \subseteq E_1$, the measurability of E_1 implis that
$$\varphi(P) + \varphi(Q - E_2) = \varphi(P \cup (Q - E_2))$$
Note that $P \cup (Q - E_2) \subseteq E_2^c$. Since $Q \cap E_2 \subseteq E_2$, we have
$$\varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) = \varphi(P \cup Q)$$
Finally , we have
$$\begin{aligned} \varphi(P) + \varphi(Q) &= \varphi(P) + \varphi(Q \cap E_2) + \varphi(Q - E_2) \\ &= \varphi(P \cup (Q - E_2)) + \varphi(Q \cap E_2) \\ &= \varphi(P \cup Q) \end{aligned}$$
4. Let $S_k = \bigcup_{i=1}^{k} E_i$. We proceed by finite induction and first note that the result is obviously true for $k = 1$. For $k > 1$ assume that S_k is φ-measurable and that
$$\varphi(A) \geq \sum_{i=1}^{k} \varphi(A \cap E_i) + \varphi\left(A \cap S_k^c\right), \quad \forall A \subseteq X$$

Then

$$\begin{aligned}\varphi(A) &= \varphi(A\cap E_{k+1}) + \varphi\left(A\cap E_{k+1}^c\right)\\ &= \varphi(A\cap E_{k+1}) + \varphi\left(A\cap E_{k+1}^c\cap S_k^c\right) + \varphi\left(A\cap E_{k+1}^c\cap S_k\right)\\ &= \varphi(A\cap E_{k+1}) + \varphi\left(A\cap S_{k+1}^c\right) + \varphi(A\cap S_k)\end{aligned}$$

Replaced A by $A \cap S_k$, we have

$$\varphi(A\cap S_k) \geq \sum_{i=1}^{k}\varphi(A\cap S_k\cap E_i) + \varphi\left(A\cap S_k\cap S_k^c\right) = \sum_{i=1}^{k}\varphi(A\cap E_i)$$

Thus

$$\varphi(A) \geq \sum_{i=1}^{k+1}\varphi(A\cap E_{k+1}) + \varphi\left(A\cap S_{k+1}^c\right)$$

Then the subadditivity gives that

$$\varphi(A) \geq \varphi(A\cap S_{k+1}) + \varphi\left(A\cap S_{k+1}^c\right)$$

this , inturn , implies that S_{k+1} is φ-measurable. Since we now know for every set $A \subseteq X$ and for all positive integers k

$$\varphi(A) \geq \sum_{i=1}^{k}\varphi(A\cap E_i) + \varphi\left(A\cap S_k^c\right)$$

We have

$$\varphi(A) \geq \sum_{i=1}^{\infty}\varphi(A\cap E_i) + \varphi(A\cap S^c) \geq \varphi(A\cap S) + \varphi(A\cap S^c)$$

where $S = \bigcup_{i=1}^{\infty} E_k$. Thus S is measurable . Replace A by S above , we have

$$\varphi(S) \geq \sum_{i=1}^{\infty}\varphi(S\cap E_i) + \varphi(S\cap S^c) = \sum_{i=1}^{\infty}\varphi(E_i)$$

Then the second part follows from the subadditivity of φ .

□